Recall that random variables X and Y are independent if and only if

$$P(X=x \mid Y=y) = P(X=x)$$

for all x and y, provided $P(Y=y) \neq 0$.

(This implies that

$$P(X=x, Y=y) = P(X=x) \cdot P(Y=y)$$

for all x and y.)

Conditional Independence

Let $\{X_n\}$ be a Markov chain.

We have, for example, that

$$P(X_4 = 1 \mid X_3 = 1, X_2 = 2) = P(X_4 = 1 \mid X_3 = 1)$$

<u>Question</u>: Does this mean that X_4 is independent of X_2?

<u>Answer</u>: No. They are <u>conditionally independent</u> given X_3.

We can specifically use the Markov property to show that

$$P(X_4 = x_4, X_2 = x_2 \mid X_3 = x_3)$$

$$= P(X_4 = x_4 \mid X_3 = x_3) \cdot P(X_2 = x_2 \mid X_3 = x_3)$$

$$P(X_4 = x_4, X_2 = x_2 \mid X_3 = x_3)$$

$$= P(X_4 = x_4 \mid X_3 = x_3, \cancel{X_2 = x_2}) \cdot P(X_2 = x_2 \mid X_3 = x_3)$$

M.P.

$$= P(X_4 = x_4 \mid X_3 = x_3) \cdot P(X_2 = x_2 \mid X_3 = x_3) \checkmark$$

<u>Aside</u>:

Suppose that $X_1, X_2, X_3, \ldots$ are "iid" random variables.

"independent and identically distributed"

<u>Question</u>: Is $\{X_n\}$ a Markov chain?

Consider

$$P(X_{n+1}=j \mid X_n=i, X_{n-1}=i_{n-1}, \ldots, X_1=i_1)$$

$$\overset{\text{indep}}{=\!=} P(X_{n+1}=j)$$

On the other hand,

Equal!

$$P(X_{n+1}=j \mid X_n=i) \overset{\text{indep}}{=\!=} P(X_{n+1}=j)$$